



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)  
BSc CHEMISTRY**

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |                   |                    |             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | B.Sc Chemistry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                   |                    |             |
| Course Title   | <b>SOLID WASTE MANAGEMENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |                   |                    |             |
| Type of Course | <b>VALUE ADDED COURSE (VAC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |                   |                    |             |
| Semester       | <b>IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |                   |                    |             |
| Academic Level | <b>100-199</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                | -                 | -                  | 45          |
| Pre-requisites | Fundamental knowledge of chemistry<br>Chemical processes that occur in the environment, including those related to pollution<br>Basic understanding of environmental chemistry and the impact of solid waste on ecosystems and human health                                                                                                                                                                                                                                                                                                                     |                  |                   |                    |             |
| Course Summary | This course provides an overview of solid waste management principles, practices, and policies. It covers the generation, collection, transportation, and disposal of solid waste, with a focus on sustainable waste management strategies. The course includes a discussion of waste management strategies for promoting waste reduction, reuse, and recycling. Through this course, the students can explore best practices for sustainable waste management, including waste minimization, source separation, and the global standards for waste management. |                  |                   |                    |             |

**Course Outcomes (CO):**

| CO  | CO Statement                                                                                         | Cognitive Level* | Knowledge Category# | Evaluation Tools used           |
|-----|------------------------------------------------------------------------------------------------------|------------------|---------------------|---------------------------------|
| CO1 | To describe the concept of solid waste and its various components.                                   | U                | C                   | Instructor-created exams / Quiz |
| CO2 | To Explore Solid Waste Collection Systems and to compare different methods of solid waste collection | U                | F                   | Class test /Assignment / Quiz   |

|                                                                                                                                                                                                   |                                                                                                                |   |   |                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---|---|-------------------------------|
| CO3                                                                                                                                                                                               | To Comprehend waste reduction principles and waste management standards                                        | U | F | Class test /Assignment / Quiz |
| CO4                                                                                                                                                                                               | To Master the different processing techniques of solid waste                                                   | U | C | Class test /Assignment / Quiz |
| CO5                                                                                                                                                                                               | To understand the basic principles involved in the Land disposal of Solid waste, and its merits and drawbacks. | U | C | Class test /Assignment / Quiz |
| CO6                                                                                                                                                                                               | To familiarize common solid waste treatment technologies like composting, recycling, and incineration          | U | F | Lab work                      |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                                                                                |   |   |                               |

| Module    | Unit | Content                                                                                                                         | Hrs       | Mark      |
|-----------|------|---------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <b>I</b>  |      | <b>Solid waste</b>                                                                                                              | <b>9</b>  | <b>18</b> |
|           | 1    | Solid waste: Definition, overview of solid waste management, types of solid wastes                                              | 2         |           |
|           | 2    | sources of solid wastes, properties of solid wastes, Factors affecting the type and quality of waste                            | 2         |           |
|           | 3    | causes of solid waste generation, associated risks of solid wastes                                                              | 2         |           |
|           | 4    | Physical and chemical composition of municipal solid waste,                                                                     | 2         |           |
|           | 5    | hierarchy of waste management options.                                                                                          | 1         |           |
| <b>II</b> |      | <b>Collection, Transportation, and Processing of Solid waste</b>                                                                | <b>12</b> | <b>23</b> |
|           | 6    | Key components of solid waste management: Generation, storage (containers), collection                                          | 1         |           |
|           | 7    | Specialized collection programs (hazardous waste, bulky waste) and transportation (human powered, animal powered and motorized) | 2         |           |
|           | 8    | Recycling and resource recovery, layout of routes                                                                               | 1         |           |
|           | 9    | Methods of handling and processing of solid wastes: separation, screening,                                                      | <b>1</b>  |           |
|           | 10   | size reduction, densification, baling, cubing, compaction, and pelleting                                                        | <b>3</b>  |           |
|           | 11   | Waste reduction hierarchy (3R Principle - reduce, reuse, recycle).                                                              | <b>2</b>  |           |

|            |               |                                                                                                                                                        |          |           |
|------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|
|            | 12            | Compliance assessment and certification processes, Overview of waste management standards - ISO 14001, OHSAS 18001                                     | 2        |           |
| <b>III</b> |               | <b>Unit-3: Land disposal of Solid waste</b>                                                                                                            | <b>7</b> | <b>14</b> |
|            | 13            | Landfilling: Site selection criteria, landfill layout, landfill sections,                                                                              | 2        |           |
|            | 14            | Occurrence of gases and leachate in landfills: composition and characteristics,                                                                        | 2        |           |
|            | 15            | generation factors, initial adjustment phase, transition phase, acid formation phase, methane formation phase, maturation phase of gases and leachate, | 2        |           |
|            | 16            | advantages and disadvantages of Land disposal of Solid waste                                                                                           | 1        |           |
| <b>IV</b>  | <b>Unit-4</b> | <b>Composting and Thermal treatment</b>                                                                                                                | <b>8</b> | <b>15</b> |
|            | 17            | Composting: definition, types, process description, design and operational consideration of aerobic composting;                                        | 2        |           |
|            | 18            | process. Description, design and operational consideration of anaerobic composting; Vermicomposting;                                                   | 3        |           |
|            | 19            | Thermal conversion methods: incineration/combustion, pyrolysis and gasification, energy recovery system                                                | 3        |           |
| <b>V</b>   | Open Ended    | <b>Open-ended experiments - Suggestions</b><br>Biomedical and E-waste management, Case study etc.                                                      | <b>9</b> |           |

## References

1. Gupta O.P, Elements of Solid Hazardous Waste Management, Khanna Book Publishing Co., Delhi Ed. 2018
2. Bhide, A. D., Solid Waste Management, Indian National Scientific Documentation Centre, New Delhi.
3. George Techobanoglous, Kreith, Frank., Solid Waste, McGraw Hill Publication, New Delhi.
4. Sasikumar, K., Solid Waste Management, PHI learning, Delhi.
5. Hosetti, B.B., Prospect and Perspectives of Solid Waste Management, New Age International Publisher.

**Mapping of COs with PSOs and POs**

|         | PS<br>O1 | PS<br>O2 | PS<br>O3 | PS<br>O4 | PS<br>O5 | PS<br>O6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---------|----------|----------|----------|----------|----------|----------|-----|-----|-----|-----|-----|-----|-----|
| CO<br>1 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |
| CO<br>2 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |
| CO<br>3 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |
| CO<br>4 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |
| CO<br>5 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |
| CO<br>6 | 1        |          |          |          |          | 2        | 1   |     |     |     | 2   | 2   |     |

**Correlation Levels :**

| Level | Correlation        |
|-------|--------------------|
| 0     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Discussion / Seminar
- Internal Theory
- Assignments / Viva
- End Semester Exam (70%)

**Mapping of COs to Assessment Rubrics**

|     | Internal Theory | Assignment / Viva | End Semester Examination |
|-----|-----------------|-------------------|--------------------------|
| CO1 | ✓               | ✓                 | ✓                        |
| CO2 | ✓               | ✓                 | ✓                        |
| CO3 | ✓               | ✓                 | ✓                        |
| CO4 | ✓               | ✓                 | ✓                        |
| CO5 | ✓               | ✓                 | ✓                        |
| CO6 | ✓               | ✓                 | ✓                        |

## **MULTI-DISCIPLINARY COURSES**